

INSTITUTE OF ENGINEERING AND MANAGEMENT, KOLKATA

Department of Computer Science & Engineering

LABORATORY ASSIGNMENT REPORT ASSIGNMENT 7: REPORTS AND LOGS

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Course	:	Python for Automation
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1. MODULE NAME & EXPERIMENT TITLE

Module Name: Reports and Logs

Experiment Title: HTML Report Generation and Custom Logging using the Log Keyword for Debugging in Robot Framework

2. PROBLEM STATEMENT

Develop a Robot Framework test suite that focuses on test reporting and custom execution logging:

- Execute a test suite and generate HTML reports (`report.html` and `log.html`) containing detailed test case summaries and execution outcomes.
- Utilize the Log keyword to write custom log messages with different logging levels (`INFO`, `WARN`, `DEBUG`) during test execution for effective debugging.

3. OBJECTIVE

- To understand the structure of Robot Framework output artifacts (`output.xml`, `log.html`, `report.html`).
- To embed custom debugging information and diagnostic checkpoints into the execution log using the Log keyword.
- To categorize execution messages using distinct log levels (`INFO`, `WARN`, `DEBUG`) to assist in root-cause analysis when debugging failing tests.

4. TOOLS, SOFTWARE, AND CONCEPTS USED

- **Programming Language & Engine:** Python 3.11.x, Robot Framework 6.1
- **Report Engine:** Built-in Robot Framework HTML Report and Log Generator
- **Development Environment:** Visual Studio Code (VS Code)
- **Key Concepts:** Custom Test Logging (Log), Log Levels (INFO, WARN, DEBUG), Output Artifact Analysis (`log.html` & `report.html`), Execution Status Breakdown.

5. SOURCE CODE / IMPLEMENTATION STEPS

Test Suite Script (assignment7_reports.robot)

```
assignment7_reports.robot - Custom Logging & Reporting Suite  Robot Framework

1  *** Settings ***

2  Documentation Assignment 7 - Custom Debug Logging and HTML Report Generation

3

4  *** Variables ***

5  ${TEST_USER} student_tester

6  ${API_RESPONSE} 200
```

6. OUTPUT WITH CLEARLY LABELLED SCREENSHOTS

The test suite was executed from the terminal using `robot --loglevel DEBUG assignment7_reports.robot`. This ensured all custom 'INFO', 'WARN', and 'DEBUG' level messages were captured in 'log.html' and 'report.html'.

Screenshot 1: VS Code Terminal Output showing Generated Log and Report Paths

```
PS C:\Users\Ashmita\Automation_Lab> robot --loglevel DEBUG assignment7_reports.robot
===== Assignment7 Reports
===== TC_01: Verify
Logging Levels And Custom Debug Messages | PASS | -----
----- [ WARN ] High response time detected during step execution!
Execution log recorded in report.html TC_02: Warning Level Message Trigger Check | PASS | -----
----- Assignment7 Reports | PASS
| 2 tests, 2 passed, 0 failed
===== Output:
C:\Users\Ashmita\Automation_Lab\output.xml Log: C:\Users\Ashmita\Automation_Lab\log.html Report:
C:\Users\Ashmita\Automation_Lab\report.html PS C:\Users\Ashmita\Automation_Lab>
```

Screenshot 2: Custom Log Statements and Log Levels Captured inside log.html

Robot Framework HTML Log - Execution Details (log.html)

Time	Level	Message
13:14:02.102	INFO	Starting test execution for user: student_tester
13:14:02.105	INFO	Checking API Connection endpoint status...
13:14:02.108	DEBUG	Debug trace: API status code is 200
13:14:02.112	WARN	High response time detected during step execution!

7. BRIEF RESULT, OBSERVATION, AND CONCLUSION

Brief Result:

The test suite executed with status PASS across both test cases. Robot Framework generated the execution summary artifacts ('output.xml', 'log.html', and 'report.html'). All custom messages logged with Log were accurately categorized under 'INFO', 'WARN', and 'DEBUG' levels within the interactive 'log.html' document.

Observation:

- The Log keyword provides customized debugging messages directly into the HTML log file without cluttering standard terminal output.
- Setting appropriate log levels (e.g., '--loglevel DEBUG') enables deep variable inspection during debugging while allowing clean high-level output during standard production test runs.
- 'WARN' level logs automatically surface in both terminal summary and log header sections, alerting testers to potential bottlenecks or non-fatal anomalies.

Conclusion:

In this experiment, I learned how to generate and analyze Robot Framework HTML reports and logs, as well as how to write custom debugging messages using the Log keyword. This facilitates efficient post-execution troubleshooting and clear test reporting.